St. George's University
Grenada, West Indies

Basic Principles of Medicine 1

Module: Musculoskeletal System | Lecture 10

Clinical Anatomy of the Thorax

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Recommended Reading

- Drake, Vogl, Mitchell. Gray's Anatomy for students. 4th Ed. Elsevier.
- Printed Textbook: Chapter 3 pg 140-168
- Online Textbook: Chapter 3
  - Conceptual Overview
  - Regional Anatomy
    - Pectoral region
    - Thoracic wall
    - Diaphragm
    - Movements of the thoracic wall and diaphragm during breathing
    - Pleural cavities
- Pay special attention to the green "In the Clinic" boxes for clinical correlations related to this topic

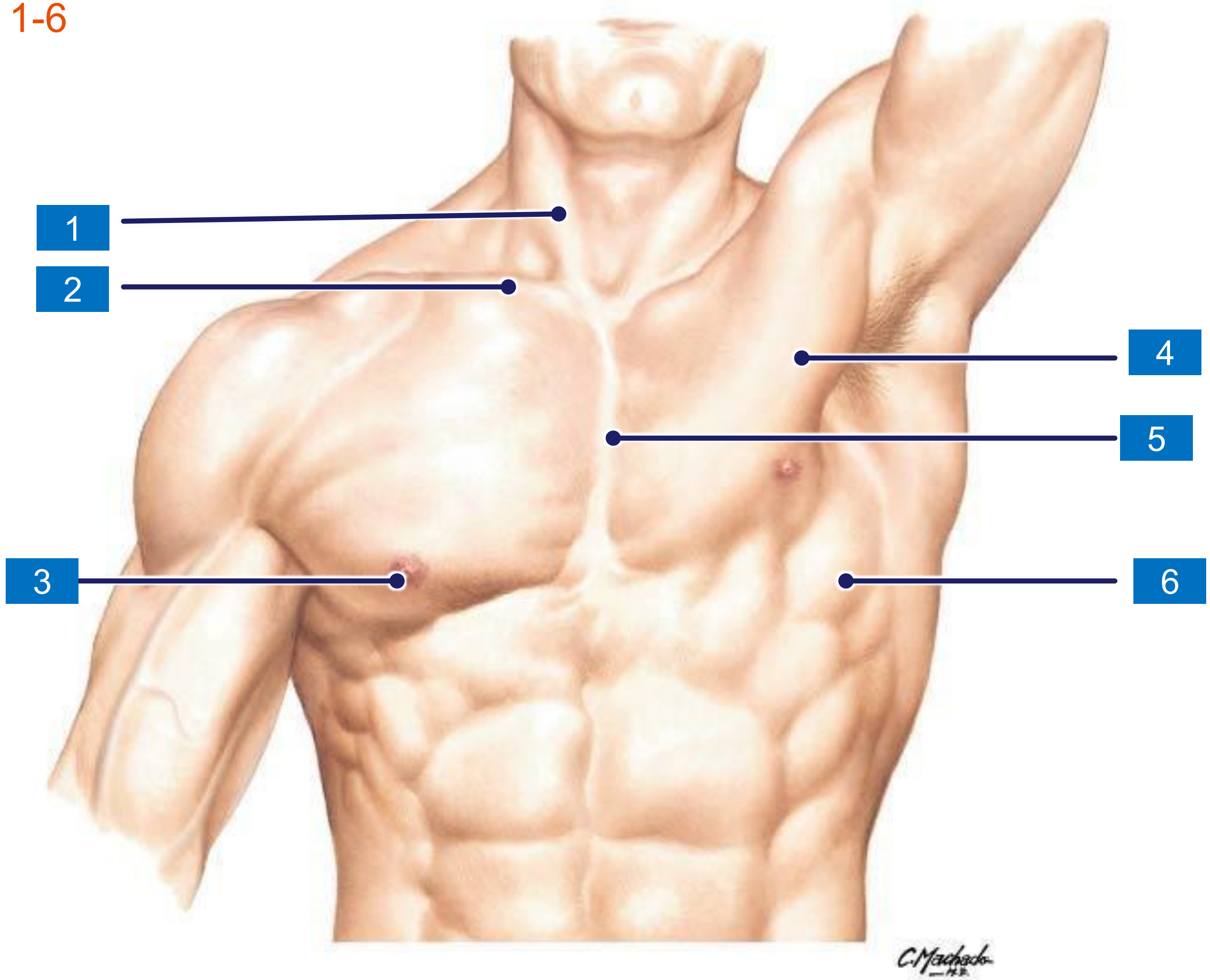
OBJECTIVES

SOM.MKII.BPM1.1.MSK.1.ANAT.1226	Explain the relationship between dermatomes of the thoracic wall and the emergence/appearance of herpes zoster (shingles).
SOM.MKII.BPM1.1.MSK.1.ANAT.1227	Describe the “milk line” and explain its clinical significance.
SOM.MKII.BPM1.1.MSK.1.ANAT.1228	Describe polythelia, polymastia, amastia, gynecomastia and their clinical significance.
SOM.MKII.BPM1.1.MSK.1.ANAT.1229	Explain the clinical importance of the lymphatic drainage of a cancerous mammary gland.
SOM.MKII.BPM1.1.MSK.1.ANAT.1230	Describe the anatomy and the clinical importance of the retro mammary space and the axillary tail of Spence.
SOM.MKII.BPM1.1.MSK.1.ANAT.1231	Describe the anatomical basis for the visible signs of breast cancer including peau d‘orange, skin dimpling and nipple retraction.
SOM.MKII.BPM1.1.MSK.1.ANAT.1232	Describe a sentinel lymph node and its clinical significance.
SOM.MKII.BPM1.1.MSK.1.ANAT.1233	Describe the neurovascular structures that are at risk during a radical mastectomy.
SOM.MKII.BPM1.1.MSK.1.ANAT.1234	Describe lymphedema and how it can be a consequence of radical mastectomy with lymphatic clearance.
SOM.MK..BPM1.1.MSK.1.ANAT.1239	Define pleurisy (pleuritis) and explain why it causes localized (somatic) pain
SOM.MK..BPM1.1.MSK.1.ANAT.1240	Describe thoracocentesis and explain why the needle is passed in the lower border of the intercostal space
SOM.MKII.BPM1.1.MSK.1.ANAT.1241	Describe the ribs levels at the midclavicular, midaxillary and paravertebral lines where a thoracocentesis can be performed

Thorax

Surface anatomy review

- Label the structures 1-6

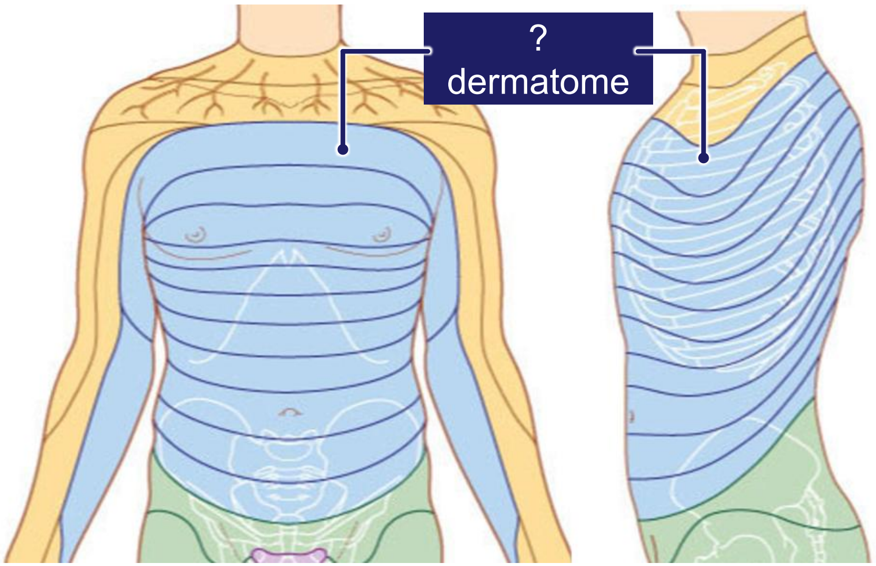


Dermatomes of the thoracic wall

- Pain from the thoracic wall structures including the pleura is felt on the skin
- The pain is mediated by **somatic sensory nerves** and as a result is felt at the corresponding area
- Each thoracic spinal nerve (T1–T12) supplies a **strip of skin (dermatome)** at roughly the same level as its rib.
- **Clinical Causes of Thoracic Wall Pain**
  - **Musculoskeletal**: Inflammation of costochondral joints or costovertebral joints.
  - **Pleurisy**:
    - If **parietal pleura** is affected: pain is localized to specific dermatomes.
    - Central diaphragmatic & mediastinal pleura → pain via **phrenic nerve (C3–C5)** → referred pain to shoulder.
    - Costal pleura → pain at **corresponding rib level dermatome**.
  - **Skin lesions**: e.g., **Herpes zoster (shingles)** → dermatomal distribution rash.

Innervation of the thoracic wall

- **Posterior rami** → supply skin of the back.
- **Ventral rami** → supply anterior & lateral thoracic wall (via anterior and lateral cutaneous branches).



Varicella Zoster

- **Pathophysiology**
  - After **chickenpox (primary infection)**, the **varicella zoster virus** does not leave the body completely.
  - It remains **dormant** in **sensory ganglia**:
  - Dorsal root ganglia of spinal nerves
  - Trigeminal ganglion (cranial nerve V)
  - Geniculate ganglion of the facial nerve (CN VII)
  - Under conditions like **stress, immunosuppression, or steroids**, the virus may **reactivate**.

- **Clinical Features**
  - **Painful, burning sensation** precedes rash.
  - Rash appears as **clusters of vesicles on an erythematous base**.
  - Distribution is **dermatomal** (follows sensory nerve distribution).
  - Classically **unilateral**, does not cross the midline.
  - Most common sites: thoracic dermatomes and trigeminal nerve distribution (ophthalmic branch is especially concerning).

- **Pain is mediated by the intercostal nerves (thoracic) or cranial nerve sensory branches (depending on site).**

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The Milk line (ridges)

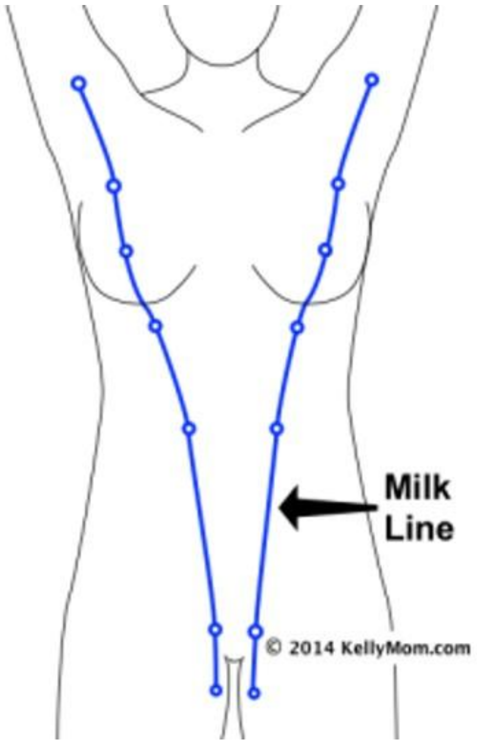
- During embryonic development, there are **two vertical thickenings of ectoderm** running from the **axilla → inguinal region**.
- These are called the **milk lines** or **mammary ridges**.

◆ **Normal Development**

- In humans, **only one pair** of sites (in the thoracic region) normally differentiates into **mammary tissue (breasts + nipples)**.
- The **rest of the milk line involutes** (regresses).

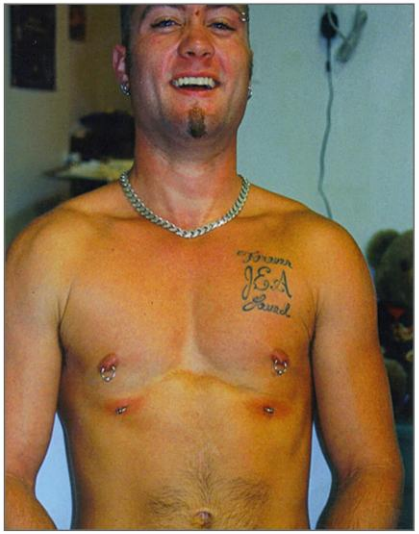
◆ **Abnormalities**

- If parts of the milk line fail to regress, remnants may persist:
  - **Polythelia** → extra nipples (may look like moles).
  - **Polymastia** → extra breast tissue.
- These can occur **anywhere along the milk line** (from axilla to groin).
- Ectopic breast tissue can be **functional** (enlarge during pregnancy, lactation) and may also develop **breast disease (e.g., carcinoma)**.



Mammary gland-Anomalies of development

<u>Polymastia:</u>
ectopic mammary gland, fully functional

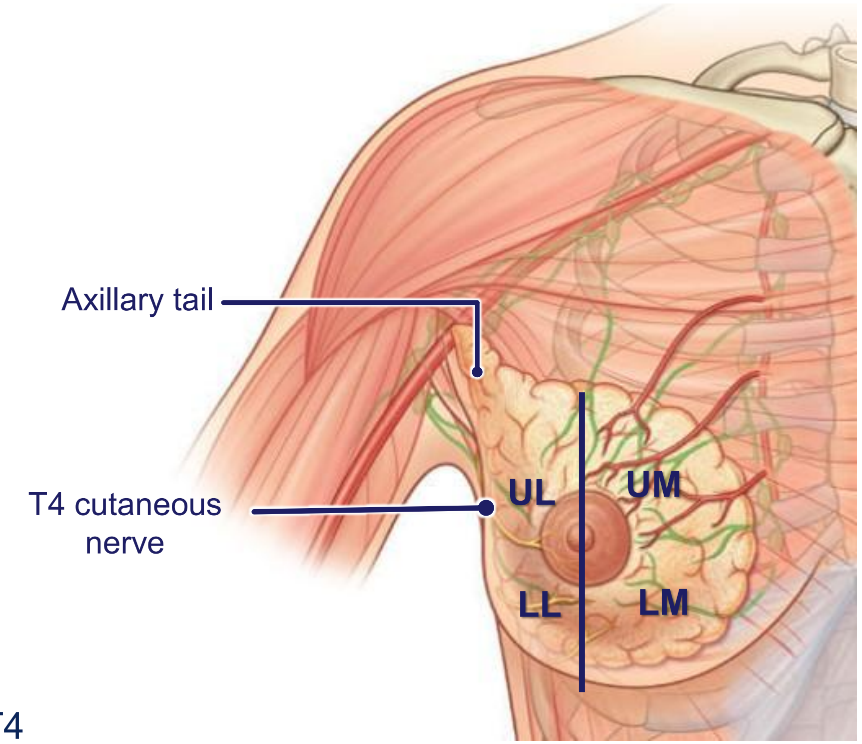


<u>Polythelia:</u>
extra nipples more common in <u>males</u>

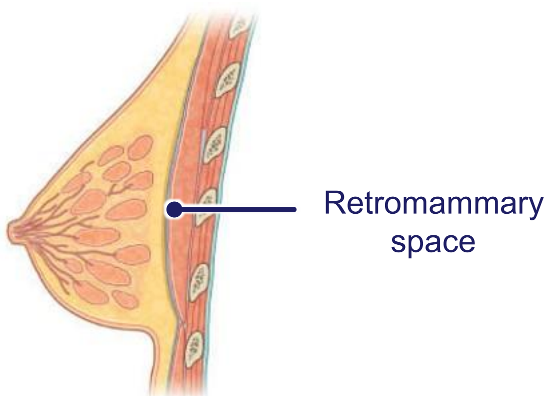
<u>Gynecomastia</u>:
Enlargement of the mammary glands in <u>males</u>

Mammary gland

- **Clinical Quadrants of the Breast**
  - **UL** = Upper lateral
  - **UM** = Upper medial
  - **LL** = Lower lateral
  - **LM** = Lower medial
- **Significance**: Quadrant classification is essential in describing tumor location and understanding **lymphatic spread & cancer metastasis**.
- Most cancers occur in the **upper lateral quadrant (UL)** → richest lymphatic drainage.



- **Axillary Tail of Spence**
- Extension of breast tissue into the **axilla**.
- **Clinical importance**: Tumors can arise here, sometimes mistaken for enlarged lymph nodes or missed if rest of breast looks normal.
- **T4 Cutaneous Nerve**
- Nipple is innervated by the **4th intercostal nerve** (T4 dermatome). This holds true even in pendulous breasts



- **Retromammary Space**
- Loose connective tissue space between the breast and pectoralis major fascia.
- Allows **mobility of breast** over chest wall.
- **Clinical significance**: In cancer, invasion into fascia causes **fixation / immobility of breast**.

Peau d' orange "orange peel"

- **Mechanism:**
- Cancer cells infiltrate and **block dermal lymphatic drainage**.
- This leads to **lymphatic edema of the skin**, giving it a **pitted appearance**, similar to the surface of an orange.
- **Clinical features:**
- **Early stage:** pitting, thickened skin, edema.
- **Advanced stage:** tumor invasion of **Cooper's suspensory ligaments**, causing **skin tethering** and **nipple retraction**.
- **Significance:**
- Strongly associated with **inflammatory breast carcinoma** or advanced **invasive ductal carcinoma**.
- Indicates aggressive disease and poorer prognosis.

Visible indications of breast cancer

These changes are only seen in advanced cases

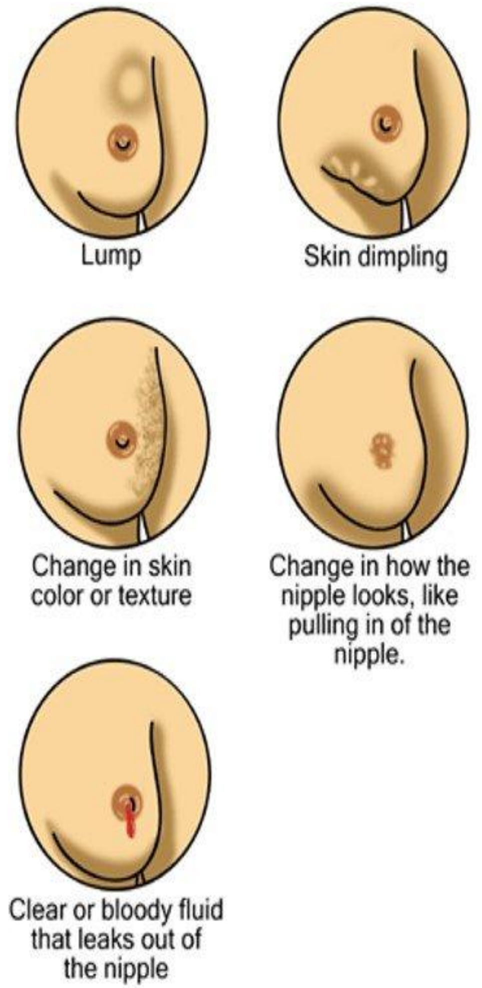
https://sites.google.com/site/breastproblems/breast-cancer/symptoms

1. **Inverted Nipple**
- Caused by **cancer infiltration of lactiferous ducts and fibrous tissue**.
- Nipple is pulled inward due to fibrosis and tethering.

2. **Dimpled Breast (Skin Dimpling)**
- Due to **involvement of Cooper's suspensory ligaments** by tumor.
- Skin appears puckered or dimpled.

3. **Discolored Breast / Skin Changes**
- Redness, thickening, or peau d'orange appearance.
- Due to **lymphatic obstruction → edema and inflammation**.

Visible indications of breast cancer



4. **Palpable Lump**
   - Hard, irregular, immobile mass.
   - Often upper outer quadrant.
5. **Skin Dimpling (as seen in diagram)**
   - Local tethering of the skin due to fibrous invasion.
6. **Change in Skin Color/Texture**
   - Can present as erythema, thickening, or peau d’orange.
7. **Change in Nipple Appearance (retraction or pulling in)**
   - Suggests invasion of ducts and fibrous tissue.
8. **Nipple Discharge (Clear or Bloody)**
   - Can be due to intraductal carcinoma or papilloma.

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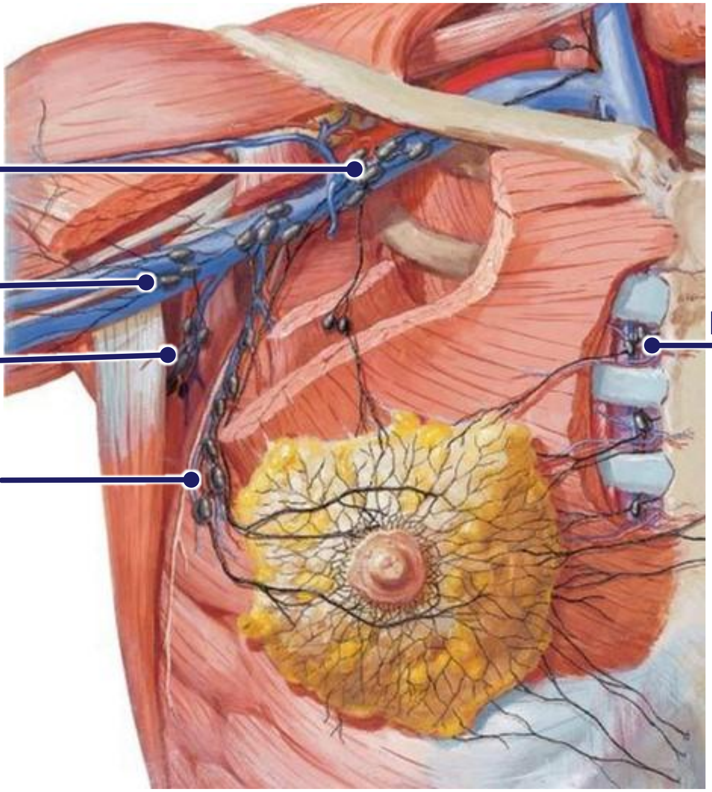
Lymphatic drainage-cancerous mammary gland

- **Carcinomas of the breast usually arise in the upper outer quadrants**
- **Primary drainage**
  - **Most lymph (≈ 75%)** drains into the **axillary lymph nodes**, especially the **anterior (pectoral) group**.
- From there, lymph spreads **node to node**:
  - **Anterior (pectoral) nodes → Central nodes → Apical nodes**
  - Other groups:
  - **Lateral (humeral) nodes, Posterior (subscapular) nodes.**
- **Medial drainage**
  - Some lymph (especially from the medial quadrants) drains to the **parasternal (internal thoracic) nodes**.
  - These nodes are important for **contralateral breast spread**.
- **Inferior drainage**
  - Small portion may drain to **subdiaphragmatic and abdominal nodes** (esp. inferior quadrants).
- **Final drainage**
  - Lymph eventually drains into the **bronchomediastinal and subclavian trunks**, then into the **venous system** via the **thoracic duct (left) or right lymphatic duct (right)**.



Sentinel Node

Defined as: **the first node into which lymph drains from a primary tumor**.

Sentinel node biopsy is crucial in **breast cancer staging**, because if this node is free of cancer, further axillary dissection may be avoided.

Lymphedema

➢Causes

1. **Surgical removal of axillary lymph nodes** (axillary dissection) in breast cancer surgery.
2. **Radiation therapy** damaging lymphatic channels.
3. **Trauma or infection** obstructing lymphatic vessels.

➢Pathophysiology

- Axillary lymph nodes are critical for **lymphatic drainage of the upper limb**.
- When they are removed or damaged, lymph cannot effectively return to the venous system.
- This leads to **lymphatic stasis → interstitial fluid accumulation → chronic swelling** of the arm.

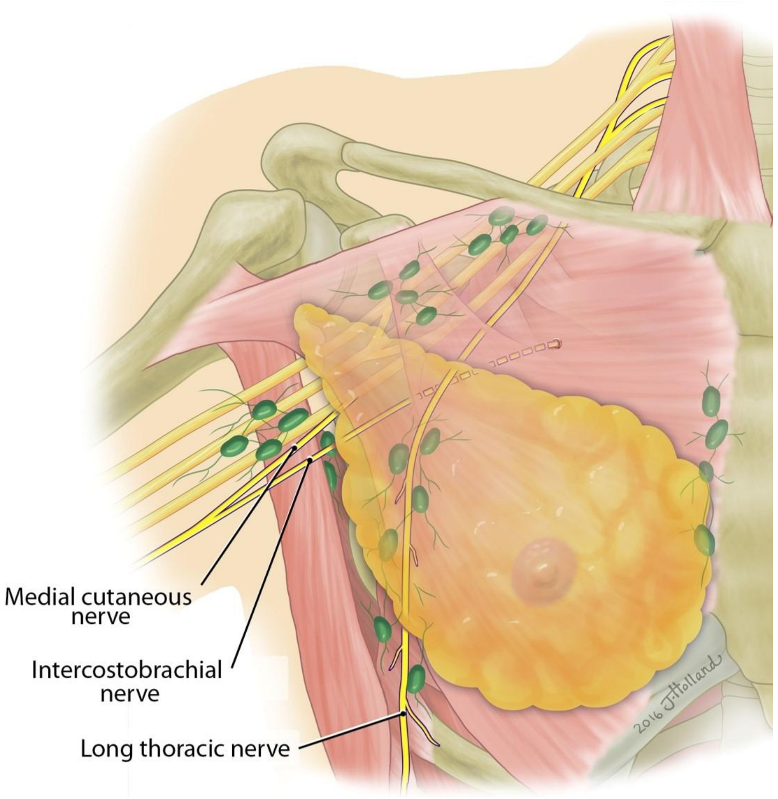
➢Clinical Features

- **Chronic, non-pitting edema** of the arm and hand.
- Heaviness, discomfort, limited mobility.
- Skin changes: fibrosis, thickening, sometimes infections (cellulitis).

Structures at risk during radical mastectomy

- Surgical resection of the entire breast
- It includes removal of the axillary tail and associated lymph nodes
  - The number and extent of lymph node removal is determined by the stage (axillary clearance)
- Three nerves are commonly damaged
  - Long thoracic
  - Thoracodorsal
  - Intercostobrachial

  Consequences?
- Damage occurs during lymph node removal
  - May also occur due to scarring
  - Causes muscle weakness, pain and restricts movement



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Pleurisy/Pleuritis

INFLAMMATION OF THE PLEURA

The inflamed pleural layers rub against each other every time the lungs expand

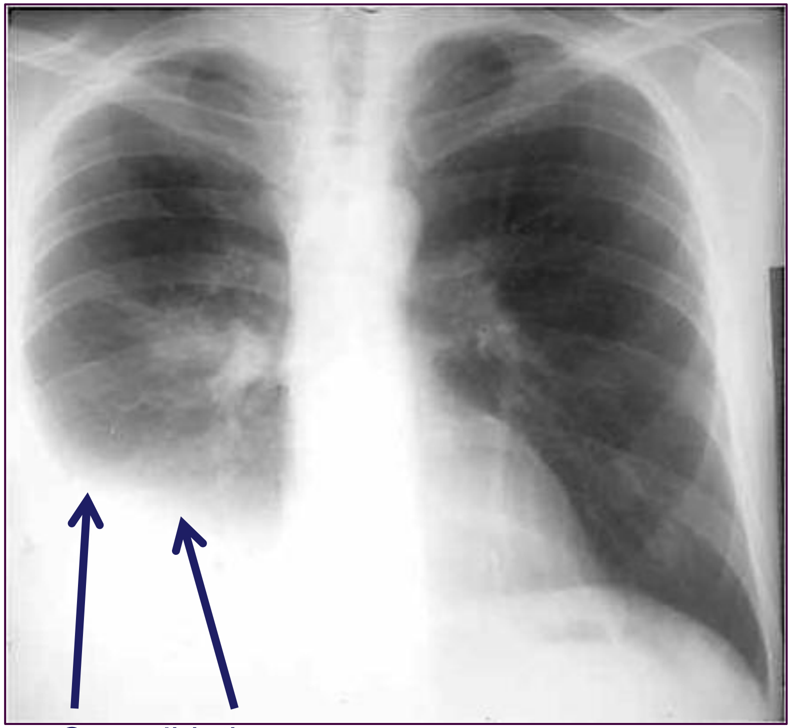
Causing **sharp pain** during respiration

Pain is localized to the area affected because it is supplied by **somatic nerves**

May radiate to the shoulder due to the phrenic nerve innervation to the central diaphragmatic pleura

On auscultation pleural friction rub (scratching) sounds are evident as long as the patient breathes

Pleural frictional rub



Consolidation

Chest Procedures

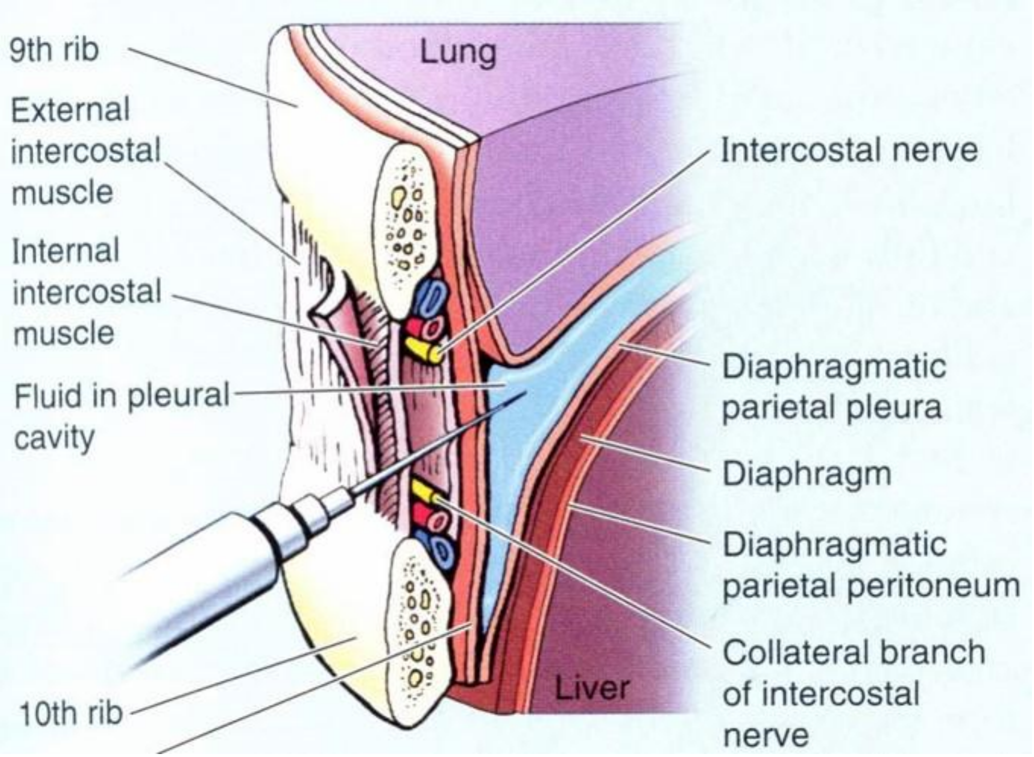
Thoracostomy

- Chest tube inserted to drain fluid or air from pleural cavity

Thoracocentesis

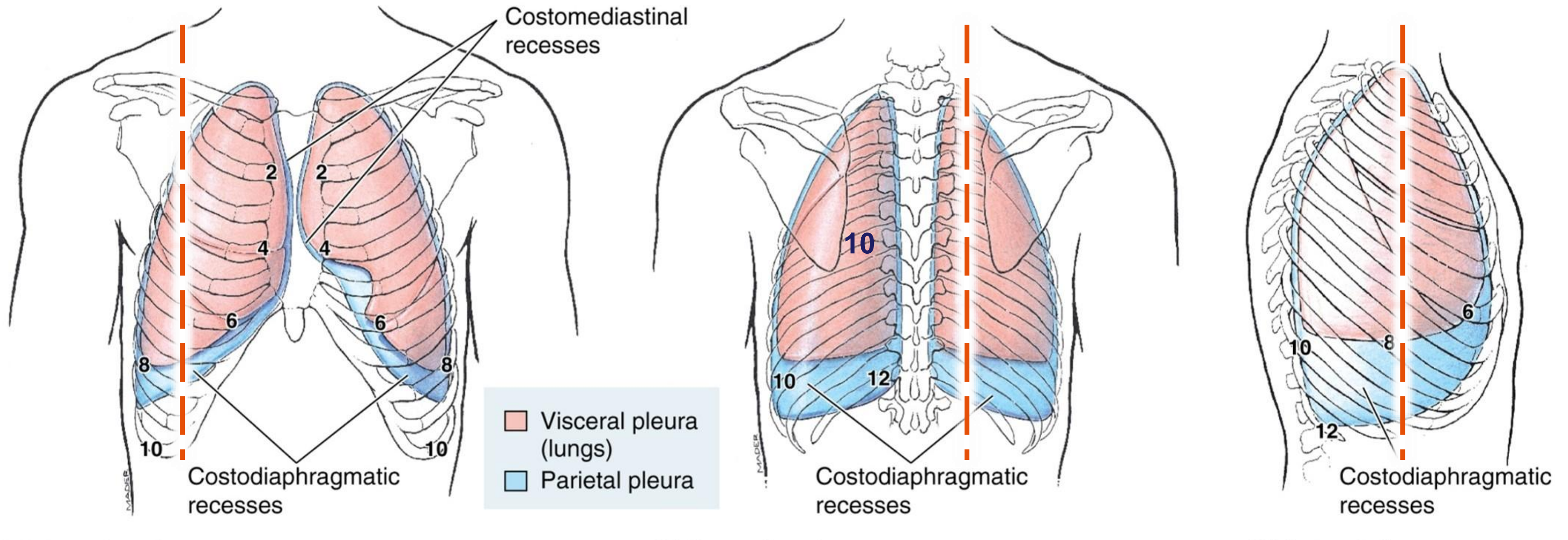
- “pleural tap” to sample fluid from pleural space

- Needle inserted into the pleural cavity to obtain a sample of fluid.
- **9th ICS →midaxillary line during expiration**, directed slightly upwards.
- Collateral branches are of no clinical significance, so needle is always superior to the rib



Surface landmarks for pleura

Between these points is the costo-diaphragmatic recess
(ideal site for accessing pleural cavity - thoracocentesis etc.)



(B) Anterior view

(C) Posterior view

(D) Lateral view

*At rest (exhaled)	Inferior border of lung	Inferior edge of parietal pleura
Mid-clavicular line	Rib 6	Rib 8
Mid-axillary line	Rib 8	Rib 10
Paravertebral	TV10	TV12